

COCA Preprocessing Pipeline- Development Report

This report summarizes the end-to-end development of a high-throughput preprocessing pipeline designed to standardize 1,155 cardiac CT scans for the PREDICT project evaluation.

Phase 1: Directory Flattening & Discovery

The raw COCA dataset contained inconsistently nested DICOM files.

- **Implementation:** Developed `unnester.py` to recursively traverse numeric patient folders and move `.dcm` files to the top-level directory.
- **Outcome:** Resolved "missing series" errors and established a flat hierarchy required for reliable volume discovery by SimpleITK.

Phase 2: Canonical Conversion & XML Ground-Truth Extraction

- **Logic:** Engineered `COCA_processor.py` to convert DICOM series to compressed NIfTI volumes (`.nii.gz`) using stable 12-character hex IDs.
- **Path Correction:** Initially encountered a "zero-voxel" error due to misaligned label paths. Corrected the `xml_root` to `Gated_release_final/calcium_xml` to successfully parse official Stanford `.plist` files.
- **Result:** Generated 3D binary masks and extracted accurate voxel-level calcium counts for 1,155 scans.

Phase 3: Spatial Standardization (Anisotropic Resampling)

- **Logic:** Implemented `COCA_resampler.py` to standardize all volumes to $0.7 \times 0.7 \times 3.0$ mm³ resolution.
- **Justification:** This configuration preserves the native 3.0 mm axial slice thickness to avoid partial volume artifacts and cross-slice blurring, while maintaining sub-millimeter in-plane resolution for calcification detection.

Phase 4: Stratified Cohort Partitioning

- **Logic:** Developed `split_data.py` to address the inherent class imbalance of the COCA dataset.
- **Method:** Categorized scans into three strata based on voxel burden: **Null** (0), **Emergent** (<500), and **Established** (≥500).
- **Outcome:** Executed a 70/15/15 split that maintains a consistent ~42% zero-calcium ratio across all sets to prevent model bias.

Phase 5: Project-Specific Preprocessing & Data Loading

Integrated clinical priors into the MONAI-based loading pipeline within `project1_task.py`:

- **HU Windowing:** Applied a window of `[-150, 500]` to isolate cardiac tissues and calcified plaque.

- **Balanced Sampling:** Utilized RandCropByPosNegLabeld with a 1:1 ratio to ensure foreground representation in 3D patches.
- **Augmentation:** Implemented Gaussian Noise, Random Rotation, and Contrast adjustment to improve model robustness.

Phase 6: Final Statistical Analysis

Validated the final cohort using get_stats.py to ensure medical and technical consistency.

Cohort	Total Scans	Zero Calcium Cases (%)	Avg Calcium Burden
TRAIN	714	306 (42.9%)	607.19 voxels
VAL	217	90 (41.5%)	621.07 voxels
TEST	224	87 (38.8%)	630.06 voxels

Summary: The pipeline successfully transforms raw clinical data into a standardized, balanced, and clinically indexed research cohort, fulfilling the Common Task requirements for scaling and usability.